Introduction to Modeling for Formal Verification

Today's goals

Module 1
Introduction to Model Checking

Motivation of Verification

- Course Updates and Accessibility Support
 Reading • 1 min
- Non-Credit Students: Welcome and Where to Find Help
 Reading • 10 min
- Motivating Examples
 Video • 7 min
- High Profile Previous Bugs and Their Impacts
 Video • 10 min
- Principles of Model Checking - Chapter 1, Introduction - Part 1
 Reading • 15 min
- Hardware and Software Bugs - Wikipedia
 Reading • 20 min
- Motivation of Verification Quiz
 Graded Assignment • Grade: 100%

Overview of Verification

- Overview of Functional Verification
 Video • 5 min
- Techniques for Functional Verification
 Video • 10 min
- **Functional Verification**
 Reading • 30 min
- Principles of Model Checking - Chapter 1, Introduction - Part 2
 Reading • 15 min
- Overview of Verification Quiz
 Graded Assignment • 10 min

Formal Verification and Model Checking

- Overview of Formal Verification and Model Checking
 Video • 7 min
- Elements of Model Checking
 Video • 6 min
- Challenge to Model Checking: State Space Explosion
 Video • 5 min
- Principles of Model Checking - Section 1.1, 1.2
 Reading • 30 min
- Formal Verification and Model Checking - Wikipedia
 Reading • 30 min
- Formal Verification and Model Checking Quiz
 Graded Assignment • 10 min

Module 2
Modeling

Module 3
Modeling System Composition

Functional Verification

0:03 / 0:41 2x

Functional Verification - Wikipedia

The Wikipedia page offers additional information about functional verification. It also contains links to other wiki pages for information about different techniques for functional verification.

Read about functional verification at the following link: Functional Verification - Wikipedia Page (external link; opens in new tab).

The Quest for Bugs: Dilemmas of Hardware Verification

This article offers a deeper look into the challenges of functional verification for hardware designs.

Read about these challenges at the following link: The Quest for Bugs: Dilemmas of Hardware Verification (external link; opens in new tab).

Go to next item ✓ Completed

Like Dislike Report an issue

Go to next item →